

**AN EXPLICIT USL MATERIAL POINT  
METHOD  
IMPLEMENTATION OF  
ANISOTROPIC TWO-FIELD PHASE-FIELD  
FRACTURE OF 3D-PRINTED LAYERED  
MATERIALS**

Technical Report

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Adapted from the variational anisotropic elastoplastic phase-field model of P. Li, J. Yvonnet, C. Combescure, H. Makich, M. Nouari (*Comput. Methods Appl. Mech. Engrg.* 386 (2021) 114086), restricted to linear elasticity and recast as an explicit USL-MPM scheme

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ATHENS, JUNE 2026

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# 1 Introduction and Scope

This report extends the explicit *Update–Stress–Last* (USL) Material Point Method (MPM) phase-field fracture scheme of the companion report to the *anisotropic, two-field* fracture model of Li *et al.* for 3D printed layered materials. Two scalar damage variables are now carried: a **bulk layer damage**  $d(\mathbf{x}) \in [0, 1]$ , which has *no* preferential direction and describes cracking inside the printed layers, and a **micro interfacial damage** variable  $\alpha(\mathbf{x}) \in [0, 1]$ , which describes the progressive failure of the weak inter-layer interphases and is therefore intrinsically *anisotropic* (oriented by the layer planes). Both fields  $d = 0, \alpha = 0$  denote intact material;  $d = 1, \alpha = 1$  denote fully broken bulk and fully debonded interphase, respectively.

Three deliberate simplifications and adaptations are made relative to the source paper:

1. **Linear elasticity only.** The plastic strain energy  $\psi^p$ , the equivalent plastic strain  $p$ , the return-mapping algorithm and the von Mises criterion of the original model are *dropped*. We set  $\varepsilon^p = \mathbf{0}$ , so that  $\varepsilon^e = \varepsilon$  and the elastic energy is simply  $\psi^e = \frac{1}{2} \varepsilon : \mathbb{C}(\alpha, d) : \varepsilon$ .
2. **Homogenisation taken as given.** The construction of the effective interphase stiffness by computational homogenisation (Section 3 of the source paper) is taken for granted. We use directly the *fitted* effective matrix  $\bar{\mathbb{C}}'(\alpha)$  (the matrix form of  $\bar{\mathbb{C}}^\alpha(\alpha)$  in the layer basis), in the quadratic form  $\bar{C}'_{ij}(\alpha) = a_1^{ij} + a_2^{ij}(1 - \alpha)^2$ .
3. **Explicit, dissipative phase fields.** Exactly as in the companion report, each elliptic damage stationarity condition is turned into a parabolic (Allen–Cahn type) evolution law by adding a dissipative micro-force. Here *two* viscosities are introduced,  $\eta_d$  and  $\eta_\alpha$ , so that *both*  $d$  and  $\alpha$  can be marched forward in time by an explicit forward–Euler update on lumped pseudo-masses, in lock-step with the explicit momentum update.

The report is organised as follows. Section 2 defines the two damage fields, the layer orientation and the projector. Section 3 gives the energy density and the damage-dependent constitutive law. Section 4 works through the rotation of the fitted stiffness  $\bar{\mathbb{C}}'(\alpha)$  by the layer angle  $\theta$  and the construction of the anisotropic gradient tensor  $\omega^\alpha$ . Section 5 adds the two dissipative micro-forces and derives the coupled strong forms. Section 6 treats irreversibility (history  $\mathcal{H}$  for  $d$ , pointwise max for  $\alpha$ ), the weak forms and the *explicit nodal updates* of  $d$  and  $\alpha$ . Section 7 embeds the two updates in the explicit USL–MPM cycle and states the complete algorithm. Section 8 derives the *three* time-step limits: the elastodynamic CFL of the momentum update and the two parabolic limits of the  $d$  and  $\alpha$  updates.

## 2 Damage variables and layer kinematics

Consider a layered solid occupying a domain  $\Omega \subset \mathbb{R}^2$  (plane stress assumed throughout, consistent with the fitted plane-stress stiffness below) with boundary  $\partial\Omega = \partial\Omega_u \cup \partial\Omega_F$ ,  $\partial\Omega_u \cap \partial\Omega_F = \emptyset$ . Displacements  $\bar{\mathbf{u}}$  are prescribed on  $\partial\Omega_u$  and tractions  $\bar{\mathbf{F}}$  on  $\partial\Omega_F$ . All layers are assumed parallel. The mechanical state is described by the displacement  $\mathbf{u}(\mathbf{x})$ , the bulk layer damage  $d(\mathbf{x})$  and the micro interfacial damage  $\alpha(\mathbf{x})$ .

**Layer orientation and projector.** Let  $\theta$  be the angle between the layer direction  $\mathbf{e}'_1$  and the global axis  $\mathbf{e}_1$  (Fig. 3(a)–(b) of the source paper). The orthonormal *layer frame* is

$$\mathbf{e}'_1 = \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix}, \quad \mathbf{e}'_2 = \mathbf{n}^\alpha = \begin{bmatrix} -\sin \theta \\ \cos \theta \end{bmatrix}, \quad (1)$$

where  $\mathbf{n}^\alpha = \mathbf{e}'_2$  is the unit normal to the layers (the normal of the weak planes). The micro interfacial cracks are constrained to run *along* the layers, i.e. perpendicular to  $\mathbf{n}^\alpha$ . We define the projector onto the layer plane,

$$\mathbf{P}^\alpha = \mathbf{I} - \mathbf{n}^\alpha \otimes \mathbf{n}^\alpha = \mathbf{e}'_1 \otimes \mathbf{e}'_1, \quad (2)$$

so that  $\mathbf{v} = \mathbf{P}^\alpha \mathbf{u}$  is the projection of  $\mathbf{u}$  onto  $\mathbf{e}'_1$ . In components, with  $c = \cos \theta$ ,  $s = \sin \theta$ ,

$$\mathbf{n}^\alpha \otimes \mathbf{n}^\alpha = \begin{bmatrix} s^2 & -sc \\ -sc & c^2 \end{bmatrix}, \quad \mathbf{P}^\alpha = \begin{bmatrix} c^2 & sc \\ sc & s^2 \end{bmatrix}. \quad (3)$$

### 3 Energy density and damage–dependent constitutive law

In the absence of body forces and plasticity, the total energy of the cracked structure is

$$W = \int_{\Omega} (\psi^e + \psi^d + \psi^\alpha) d\Omega - \int_{\partial\Omega_F} \bar{\mathbf{F}} \cdot \mathbf{u} dS, \quad (4)$$

where  $\psi^e$  is the elastic strain energy density,  $\psi^d$  is the bulk layer fracture energy density and  $\psi^\alpha$  is the micro interfacial fracture energy density.

**Elastic energy and stress.** With  $\boldsymbol{\varepsilon} = \frac{1}{2}(\nabla \mathbf{u} + \nabla \mathbf{u}^T)$  the (small) strain and  $\boldsymbol{\varepsilon}^e = \boldsymbol{\varepsilon}$  (linear elasticity),

$$\psi^e = \frac{1}{2} \boldsymbol{\varepsilon} : \mathbb{C}(\alpha, d) : \boldsymbol{\varepsilon}, \quad \mathbb{C}(\alpha, d) = g(d) \bar{\mathbb{C}}^\alpha(\alpha), \quad g(d) = (1 - d)^2. \quad (5)$$

Here  $\bar{\mathbb{C}}^\alpha(\alpha)$  is the effective (homogenised) elasticity tensor of the interphase, expressed in the global axes, obtained by rotating the fitted stiffness  $\bar{\mathbb{C}}'(\alpha)$  (layer axes) by the layer angle  $\theta$  (Section 4). The Cauchy stress is

$$\boldsymbol{\sigma} = \frac{\partial \psi^e}{\partial \boldsymbol{\varepsilon}} = g(d) \bar{\mathbb{C}}^\alpha(\alpha) : \boldsymbol{\varepsilon} = g(d) \boldsymbol{\sigma}_0, \quad \boldsymbol{\sigma}_0 \equiv \bar{\mathbb{C}}^\alpha(\alpha) : \boldsymbol{\varepsilon}. \quad (6)$$

The stress is thus degraded by the bulk damage through  $g(d) = (1 - d)^2$ , while the interfacial damage  $\alpha$  enters through the  $\alpha$ –dependence (and  $\theta$ –rotation) of the effective stiffness  $\bar{\mathbb{C}}^\alpha(\alpha)$ .

**Bulk fracture energy (isotropic, AT2–type).** The bulk layer crack surface density and its energy are

$$\psi^d = g_c^d \gamma^d(d), \quad \gamma^d(d) = \frac{d^2}{2\ell_d} + \frac{\ell_d}{2} \nabla d \cdot \nabla d, \quad (7)$$

with  $g_c^d$  the bulk fracture toughness and  $\ell_d$  the bulk length scale. The crack density is *isotropic* (no preferred direction), consistent with bulk cracking inside the layers.

**Interfacial fracture energy (anisotropic).** The micro interfacial crack surface density is made directional through the structural tensor  $\omega^\alpha$ ,

$$\psi^\alpha = g_c^\alpha \gamma^\alpha(\alpha), \quad \gamma^\alpha(\alpha) = \frac{\alpha^2}{2\ell_\alpha} + \frac{\ell_\alpha}{2} \omega^\alpha : (\nabla\alpha \otimes \nabla\alpha), \quad (8)$$

with  $g_c^\alpha$  the interfacial toughness,  $\ell_\alpha$  the interfacial length scale, and

$$\omega^\alpha = \mathbf{I} + \xi^\alpha \mathbf{P}^\alpha, \quad \xi^\alpha \gg 1. \quad (9)$$

The scalar  $\xi^\alpha \gg 1$  heavily penalises the gradient of  $\alpha$  in the direction  $\mathbf{e}'_1$  *normal to*  $\mathbf{n}^\alpha$ , i.e. along the layers; this forces  $\alpha$  to vary slowly along the layer and rapidly across it, so that interfacial cracks smear into thin bands aligned with the weak planes. The quadratic form  $\omega^\alpha : (\nabla\alpha \otimes \nabla\alpha) = \nabla\alpha \cdot \omega^\alpha \nabla\alpha$  is the anisotropic counterpart of  $|\nabla\alpha|^2$ .

Following the source paper, the length scales are tied to the minimal element (cell) size  $h_e$  via  $\ell_d = \ell_\alpha = 2h_e$ , although they may also be regarded as material parameters.

## 4 Rotation of $\overline{\mathbf{C}}'(\alpha)$ and construction of $\omega^\alpha$ at angle $\theta$

This section is the core “geometry” of the anisotropic model: how the fitted interphase stiffness and the gradient tensor, both naturally expressed in the layer frame ( $\mathbf{e}'_1, \mathbf{e}'_2$ ), are mapped to the global frame ( $\mathbf{e}_1, \mathbf{e}_2$ ) for a layer set inclined at angle  $\theta$ .

### 4.1 The fitted effective stiffness in the layer frame

Taken as given from the computational homogenisation, the effective stiffness in the layer (primed) basis is the plane–stress  $3 \times 3$  Voigt matrix, fitted by a quadratic in  $(1 - \alpha)$ ,

$$\overline{\mathbf{C}}'_{ij}(\alpha) = a_1^{ij} + a_2^{ij} (1 - \alpha)^2, \quad \overline{\mathbf{C}}'(\alpha) = \begin{bmatrix} \overline{\mathbf{C}}'_{11} & \overline{\mathbf{C}}'_{12} & 0 \\ \overline{\mathbf{C}}'_{12} & \overline{\mathbf{C}}'_{22} & 0 \\ 0 & 0 & \overline{\mathbf{C}}'_{33} \end{bmatrix}, \quad (10)$$

in Voigt order  $\{11, 22, 12\}$  with engineering shear  $\gamma_{12} = 2\varepsilon_{12}$ . The quadratic form is chosen precisely so that the interfacial damage problem remains *linear* in  $\alpha$  (the driving term is linear in  $\alpha$ , see Section 5). As a concrete numerical instantiation, the straight interphase RVE of the source paper that was conducted for  $E = 10 \text{ GPa}$ ,  $\nu = .25$  yields:

$$\overline{\mathbf{C}}'(\alpha) = \frac{E}{1 - \nu^2} \begin{bmatrix} .8437 + .1563(1 - \alpha)^2 & \nu(1 - \alpha)^2 & 0 \\ \nu(1 - \alpha)^2 & (1 - \alpha)^2 & 0 \\ 0 & 0 & \frac{1 - \nu}{2}(1 - \alpha)^2 \end{bmatrix} [\text{GPa}], \quad (11)$$

i.e.  $a_1^{11} = .8437 \frac{E}{1 - \nu^2}$ ,  $a_2^{11} = .1563 \frac{E}{1 - \nu^2}$ ,  $a_1^{12} = a_1^{22} = a_1^{33} = 0$ ,  $a_2^{12} = \nu \frac{E}{1 - \nu^2}$ ,  $a_2^{22} = \frac{E}{1 - \nu^2}$ ,  $a_2^{33} = \frac{1 - \nu}{2} \frac{E}{1 - \nu^2}$  (GPa). Note that  $\overline{\mathbf{C}}'_{12}, \overline{\mathbf{C}}'_{22}, \overline{\mathbf{C}}'_{33} \rightarrow 0$  as  $\alpha \rightarrow 1$  (complete loss of transverse/shear

stiffness across a fully damaged interphase), while  $\bar{C}'_{11} \rightarrow a_1^{11} \neq 0$  (the layer keeps an in-plane longitudinal stiffness). Its  $\alpha$ -derivative, needed for the interfacial driving force, is

$$\frac{\partial \bar{C}'}{\partial \alpha} = \begin{bmatrix} -2a_2^{11}(1-\alpha) & -2a_2^{12}(1-\alpha) & 0 \\ -2a_2^{12}(1-\alpha) & -2a_2^{22}(1-\alpha) & 0 \\ 0 & 0 & -2a_2^{33}(1-\alpha) \end{bmatrix} = -2(1-\alpha) \mathbf{A}_2, \quad (12)$$

where  $\mathbf{A}_2$  collects the quadratic coefficients  $a_2^{ij}$ .  $\partial \bar{C}' / \partial \alpha$  is negative semi-definite, so degradation of the interphase releases energy that drives  $\alpha \rightarrow 1$ .

## 4.2 Voigt rotation to the global frame

The fourth-order rotation  $\bar{C}_{ijkl}^\alpha = R_{ip}R_{jq}R_{kr}R_{ls}\bar{C}'_{pqrs}$  of the source paper (eq. 57), with the in-plane rotation  $\mathbf{R}(\theta) = \begin{bmatrix} c & -s \\ s & c \end{bmatrix}$ , is implemented in Voigt matrix form as a congruence transformation. Define the *stress* Voigt rotation matrix

$$\mathbf{T}_\sigma(\theta) = \begin{bmatrix} c^2 & s^2 & -2cs \\ s^2 & c^2 & 2cs \\ cs & -cs & c^2 - s^2 \end{bmatrix}, \quad c = \cos \theta, \quad s = \sin \theta, \quad (13)$$

which maps a stress vector from the layer frame to the global frame,  $[\boldsymbol{\sigma}]_{\text{glob}} = \mathbf{T}_\sigma[\boldsymbol{\sigma}]_{\text{layer}}$ . The corresponding *strain* (engineering) Voigt rotation is  $\mathbf{T}_\varepsilon(\theta) = \begin{bmatrix} c^2 & s^2 & -cs \\ s^2 & c^2 & cs \\ 2cs & -2cs & c^2 - s^2 \end{bmatrix}$ , and the two satisfy the identity  $\mathbf{T}_\varepsilon^{-1}(\theta) = \mathbf{T}_\sigma^T(\theta)$ . Since  $[\boldsymbol{\sigma}]_{\text{glob}} = \mathbf{T}_\sigma[\boldsymbol{\sigma}]_{\text{layer}} = \mathbf{T}_\sigma \bar{\mathbf{C}}' [\boldsymbol{\varepsilon}]_{\text{layer}} = \mathbf{T}_\sigma \bar{\mathbf{C}}' \mathbf{T}_\varepsilon^{-1}[\boldsymbol{\varepsilon}]_{\text{glob}}$ , the effective stiffness in the global frame is the congruence

$$\boxed{\bar{\mathbf{C}}^\alpha(\alpha, \theta) = \mathbf{T}_\sigma(\theta) \bar{\mathbf{C}}'(\alpha) \mathbf{T}_\sigma^T(\theta)}. \quad (14)$$

This matrix identity is exactly equivalent to the index expression (eq. 57 of the paper) and is the form used in the code: rotate the fitted layer-frame matrix once per  $\theta$ , then it depends on  $\alpha$  only through the scalar  $(1-\alpha)^2$  multipliers inside  $\bar{\mathbf{C}}'$ . Likewise

$$\frac{\partial \bar{\mathbf{C}}^\alpha}{\partial \alpha}(\alpha, \theta) = \mathbf{T}_\sigma(\theta) \frac{\partial \bar{\mathbf{C}}'}{\partial \alpha} \mathbf{T}_\sigma^T(\theta) = -2(1-\alpha) \mathbf{T}_\sigma(\theta) \mathbf{A}_2 \mathbf{T}_\sigma^T(\theta). \quad (15)$$

**Worked component expansion.** Carrying out the product (14) for the structure of (10) (with  $\bar{C}'_{13} = \bar{C}'_{23} = 0$ ) gives the global components

$$\bar{C}_{11}^\alpha = c^4 \bar{C}'_{11} + s^4 \bar{C}'_{22} + 2c^2 s^2 \bar{C}'_{12} + 4c^2 s^2 \bar{C}'_{33}, \quad (16)$$

$$\bar{C}_{22}^\alpha = s^4 \bar{C}'_{11} + c^4 \bar{C}'_{22} + 2c^2 s^2 \bar{C}'_{12} + 4c^2 s^2 \bar{C}'_{33}, \quad (17)$$

$$\bar{C}_{12}^\alpha = c^2 s^2 \bar{C}'_{11} + c^2 s^2 \bar{C}'_{22} + (c^4 + s^4) \bar{C}'_{12} - 4c^2 s^2 \bar{C}'_{33}, \quad (18)$$

$$\bar{C}_{33}^\alpha = c^2 s^2 \bar{C}'_{11} + c^2 s^2 \bar{C}'_{22} - 2c^2 s^2 \bar{C}'_{12} + (c^2 - s^2)^2 \bar{C}'_{33}, \quad (19)$$

$$\bar{C}_{13}^\alpha = cs[c^2 \bar{C}'_{11} - s^2 \bar{C}'_{22} - (c^2 - s^2) \bar{C}'_{12} - 2(c^2 - s^2) \bar{C}'_{33}], \quad (20)$$

$$\bar{C}_{23}^\alpha = cs[s^2 \bar{C}'_{11} - c^2 \bar{C}'_{22} + (c^2 - s^2) \bar{C}'_{12} + 2(c^2 - s^2) \bar{C}'_{33}]. \quad (21)$$

The off-diagonal coupling terms  $\overline{C}_{13}^\alpha, \overline{C}_{23}^\alpha$  (normal-shear coupling) vanish at  $\theta = 0$  and  $\theta = 90^\circ$  but are nonzero for inclined layers, which is the origin of the orientation-dependent response. (For implementation one simply forms the  $3 \times 3$  product (14) numerically; the expansion is given to make the anisotropy explicit.)

### 4.3 The gradient tensor $\omega^\alpha$ at angle $\theta$

The structural tensor follows directly from the projector (3). Substituting  $\mathbf{P}^\alpha$  into (9),

$$\omega^\alpha(\theta) = \mathbf{I} + \xi^\alpha \mathbf{P}^\alpha = \begin{bmatrix} 1 + \xi^\alpha c^2 & \xi^\alpha cs \\ \xi^\alpha cs & 1 + \xi^\alpha s^2 \end{bmatrix}. \quad (22)$$

Its eigenpairs are transparent:  $\omega^\alpha \mathbf{e}'_1 = (1 + \xi^\alpha) \mathbf{e}'_1$  and  $\omega^\alpha \mathbf{n}^\alpha = 1 \cdot \mathbf{n}^\alpha$ , i.e. the eigenvalue  $1 + \xi^\alpha$  acts along the layer direction  $\mathbf{e}'_1$  and the eigenvalue 1 acts across the layers along  $\mathbf{n}^\alpha$ . The anisotropic gradient energy is then

$$\omega^\alpha : (\nabla \alpha \otimes \nabla \alpha) = \nabla \alpha \cdot \omega^\alpha \nabla \alpha = |\nabla \alpha|^2 + \xi^\alpha (\mathbf{e}'_1 \cdot \nabla \alpha)^2, \quad (23)$$

so the cost of a gradient of  $\alpha$  along the layer ( $\mathbf{e}'_1 \cdot \nabla \alpha$ ) is amplified by  $(1 + \xi^\alpha)$ . Two limiting orientations check the algebra:

$$\theta = 0^\circ : \omega^\alpha = \begin{bmatrix} 1 + \xi^\alpha & 0 \\ 0 & 1 \end{bmatrix}, \quad \theta = 90^\circ : \omega^\alpha = \begin{bmatrix} 1 & 0 \\ 0 & 1 + \xi^\alpha \end{bmatrix}, \quad (24)$$

consistent with layers aligned with  $\mathbf{e}_1$  and  $\mathbf{e}_2$ , respectively. The divergence of the associated flux, needed below, is

$$\omega^\alpha : \nabla \nabla \alpha = \nabla \cdot (\omega^\alpha \nabla \alpha) = (1 + \xi^\alpha c^2) \alpha_{,11} + 2\xi^\alpha cs \alpha_{,12} + (1 + \xi^\alpha s^2) \alpha_{,22}, \quad (25)$$

(for spatially uniform  $\theta$ , so  $\omega^\alpha$  is constant in  $\mathbf{x}$ ).

## 5 Coupled strong forms with two dissipative micro-forces

### 5.1 Energetic micro-stresses

From the energy density (4) with (5)–(9), the energetic scalar micro-stresses conjugate to  $d$  and  $\alpha$ , and the corresponding micro-stress vectors, are

$$\omega_{\text{en}}^d = \frac{\partial \psi}{\partial d} = g'(d) \psi_0^e + \frac{g_c^d}{\ell_d} d, \quad \boldsymbol{\xi}^d = \frac{\partial \psi}{\partial \nabla d} = g_c^d \ell_d \nabla d, \quad (26)$$

$$\omega_{\text{en}}^\alpha = \frac{\partial \psi}{\partial \alpha} = \frac{1}{2} g(d) \boldsymbol{\varepsilon} : \frac{\partial \overline{\mathbf{C}}^\alpha}{\partial \alpha} : \boldsymbol{\varepsilon} + \frac{g_c^\alpha}{\ell_\alpha} \alpha, \quad \boldsymbol{\xi}^\alpha = \frac{\partial \psi}{\partial \nabla \alpha} = g_c^\alpha \ell_\alpha \omega^\alpha \nabla \alpha, \quad (27)$$

where  $\psi_0^e = \frac{1}{2} \boldsymbol{\varepsilon} : \overline{\mathbf{C}}^\alpha(\alpha) : \boldsymbol{\varepsilon}$  is the elastic energy density evaluated with the (bulk-undamaged) effective stiffness, and  $g'(d) = -2(1 - d)$ .

## 5.2 Dissipative micro-forces and rate balances

Exactly as in the companion report, each phase-field balance  $\nabla \cdot \boldsymbol{\xi}^{(\cdot)} - \omega^{(\cdot)} = 0$  is rendered *parabolic* by letting the scalar micro-stress possess a dissipative part proportional to the rate of the corresponding field,

$$\omega^d = \omega_{\text{en}}^d + \eta_d \dot{d}, \quad \omega^\alpha = \omega_{\text{en}}^\alpha + \eta_\alpha \dot{\alpha}, \quad \eta_d, \eta_\alpha \geq 0, \quad (28)$$

with viscosities  $\eta_d, \eta_\alpha$  (units  $\text{Pa} \cdot \text{s}$ ). Each dissipative term satisfies the dissipation inequality ( $\eta_d \dot{d}^2 \geq 0, \eta_\alpha \dot{\alpha}^2 \geq 0$ ) and vanishes in the quasi-static limit. Substituting (26)–(28) into the two balances gives the rate forms

$$\eta_d \dot{d} = \nabla \cdot \boldsymbol{\xi}^d - \omega_{\text{en}}^d = g_c^d \ell_d \nabla^2 d - \frac{g_c^d}{\ell_d} d + 2(1-d) \mathcal{H}, \quad (29)$$

$$\eta_\alpha \dot{\alpha} = \nabla \cdot \boldsymbol{\xi}^\alpha - \omega_{\text{en}}^\alpha = g_c^\alpha \ell_\alpha \boldsymbol{\omega}^\alpha : \nabla \nabla \alpha - \frac{g_c^\alpha}{\ell_\alpha} \alpha - \frac{1}{2} g(d) \boldsymbol{\varepsilon} : \frac{\partial \bar{\mathbf{C}}^\alpha}{\partial \alpha} : \boldsymbol{\varepsilon}, \quad (30)$$

where in (29) the elastic driving energy  $\psi_0^e$  has been replaced by the irreversible history field  $\mathcal{H}$  (Section 6). Equations (29)–(30) are two Allen–Cahn type evolution laws, integrable by explicit forward Euler.

**Sign of the interfacial driving force.** Since  $\partial \bar{\mathbf{C}}^\alpha / \partial \alpha = -2(1-\alpha) \mathbf{T}_\sigma \mathbf{A}_2 \mathbf{T}_\sigma^T$  is negative semi-definite, the last term in (30) is non-negative and *drives*  $\alpha$  upward. It is convenient to define the interfacial crack driving energy

$$\mathcal{G}^\alpha := -\frac{1}{2} g(d) \boldsymbol{\varepsilon} : \frac{\partial \bar{\mathbf{C}}^\alpha}{\partial \alpha} : \boldsymbol{\varepsilon} = (1-\alpha) g(d) \boldsymbol{\varepsilon} : (\mathbf{T}_\sigma \mathbf{A}_2 \mathbf{T}_\sigma^T) : \boldsymbol{\varepsilon} \geq 0, \quad (31)$$

the  $\alpha$ -analogue of the elastic driving energy in the  $d$ -equation.

## 5.3 Summary of the coupled strong form

The complete coupled initial/boundary-value problem is

$$\rho \dot{\mathbf{v}} = \nabla \cdot [g(d) \bar{\mathbf{C}}^\alpha(\alpha) : \boldsymbol{\varepsilon}] + \rho \mathbf{b}, \quad (\text{momentum}) \quad (32a)$$

$$\eta_d \dot{d} = g_c^d \ell_d \nabla^2 d - \frac{g_c^d}{\ell_d} d + 2(1-d) \mathcal{H}, \quad (\text{bulk damage}) \quad (32b)$$

$$\eta_\alpha \dot{\alpha} = g_c^\alpha \ell_\alpha \boldsymbol{\omega}^\alpha : \nabla \nabla \alpha - \frac{g_c^\alpha}{\ell_\alpha} \alpha + \mathcal{G}^\alpha, \quad (\text{interfacial damage}) \quad (32c)$$

$$\boldsymbol{\sigma} \cdot \mathbf{n} = \bar{\mathbf{F}}, \quad \nabla d \cdot \mathbf{n} = 0, \quad \boldsymbol{\omega}^\alpha \nabla \alpha \cdot \mathbf{n} = 0, \quad (\text{boundary}) \quad (32d)$$

coupled through (i)  $g(d)$ , which degrades the stress; (ii)  $\bar{\mathbf{C}}^\alpha(\alpha)$ , whose  $\alpha$ -degradation softens the stress and supplies  $\mathcal{H}$ ; and (iii)  $\mathcal{G}^\alpha$ , through which the elastic state and the bulk damage  $g(d)$  drive the interface. In the quasi-static limit  $\eta_d, \eta_\alpha \rightarrow 0$ , (32b)–(32c) collapse to the stationarity conditions (eqs. 41 and 44 of the source paper, with  $\boldsymbol{\varepsilon}^e = \boldsymbol{\varepsilon}$ ).



## 6 Discretisation and explicit nodal updates

### 6.1 Irreversibility

Both damage fields must be non-decreasing,  $\dot{d} \geq 0$ ,  $\dot{\alpha} \geq 0$ .

**Bulk damage: history field.** As specified,  $d$  is driven by a monotone history field that records the maximum elastic energy density attained, evaluated with the  $\alpha$ -degraded effective stiffness,

$$\mathcal{H} = \max_{\tau \in [0, t]} \psi_0^e(\tau), \quad \psi_0^e = \frac{1}{2} \boldsymbol{\varepsilon} : \overline{\mathbf{C}}^\alpha(\alpha) : \boldsymbol{\varepsilon}, \quad (33)$$

with  $\dot{\mathcal{H}} \geq 0$ ,  $\psi_0^e - \mathcal{H} \leq 0$ ,  $\dot{\mathcal{H}}(\psi_0^e - \mathcal{H}) = 0$ . This is the  $\mathcal{H}$  used in (32b).

**Interfacial damage: pointwise maximum.** Following the source paper (its eq. 15), irreversibility of  $\alpha$  is enforced by a pointwise maximum applied after the explicit update,

$$\alpha^{t+\Delta t} \leftarrow \max(\alpha^t, \alpha_{\text{trial}}^{t+\Delta t}), \quad (34)$$

where  $\alpha_{\text{trial}}^{t+\Delta t}$  is the value produced by the explicit evolution (32c).

### 6.2 Weak forms

With test functions  $\delta d$ ,  $\delta \alpha$  and the natural boundary conditions (32d), multiplying (29)–(30) and integrating by parts over the reference domain  $\Omega_0$  gives

$$\int_{\Omega_0} \left\{ \eta_d \dot{d} \delta d + \frac{g_c^d}{\ell_d} d \delta d + g_c^d \ell_d \nabla d \cdot \nabla \delta d - 2(1-d) \mathcal{H} \delta d \right\} dV = 0, \quad (35)$$

$$\int_{\Omega_0} \left\{ \eta_\alpha \dot{\alpha} \delta \alpha + \frac{g_c^\alpha}{\ell_\alpha} \alpha \delta \alpha + g_c^\alpha \ell_\alpha (\boldsymbol{\omega}^\alpha \nabla \alpha) \cdot \nabla \delta \alpha + \frac{1}{2} g(d) \boldsymbol{\varepsilon} : \frac{\partial \overline{\mathbf{C}}^\alpha}{\partial \alpha} : \boldsymbol{\varepsilon} \delta \alpha \right\} dV = 0. \quad (36)$$

In each, the first term is the new dissipative contribution; the rest is the energetic residual of the original quasi-static formulation. (For reference, the implicit tangent of the  $\alpha$ -problem is linear in  $\alpha$  precisely because  $\partial \overline{\mathbf{C}}^\alpha / \partial \alpha \propto (1 - \alpha)$  and the gradient term is linear; this is the reason for the quadratic fit (10).)

### 6.3 Spatial discretisation and explicit updates

Displacement and both damage fields share the MPM background-grid shape functions  $N_I$ , with  $\mathbf{B}_I = \nabla N_I$ :

$$d = \sum_I N_I \hat{d}_I, \quad \alpha = \sum_I N_I \hat{\alpha}_I, \quad \nabla d = \sum_I \mathbf{B}_I \hat{d}_I, \quad \nabla \alpha = \sum_I \mathbf{B}_I \hat{\alpha}_I. \quad (37)$$

Discretising the rates by forward Euler,  $\dot{d} \approx (d_I^{t+\Delta t} - d_I^t)/\Delta t$ ,  $\dot{\alpha} \approx (\alpha_I^{t+\Delta t} - \alpha_I^t)/\Delta t$ , and lumping the viscous terms onto the diagonal pseudo-masses

$$m_I^d = m_I^\alpha = \int_{\Omega_0} N_I dV \xrightarrow{\text{MPM}} \sum_p N_I(\mathbf{x}_p) V_p^0, \quad (38)$$

the two nodal balances solve explicitly. Writing the *energetic* nodal residuals as

$$F_I^d = \int_{\Omega_0} \left\{ \frac{g_c^d}{\ell_d} d^t N_I + g_c^d \ell_d \mathbf{B}_I \cdot \nabla d^t - 2(1 - d^t) N_I \mathcal{H} \right\} dV, \quad (39)$$

$$F_I^\alpha = \int_{\Omega_0} \left\{ \frac{g_c^\alpha}{\ell_\alpha} \alpha^t N_I + g_c^\alpha \ell_\alpha \mathbf{B}_I \cdot (\boldsymbol{\omega}^\alpha \nabla \alpha^t) + \frac{1}{2} g(d^t) \boldsymbol{\varepsilon} : \frac{\partial \bar{\mathbf{C}}^\alpha}{\partial \alpha} : \boldsymbol{\varepsilon} N_I \right\} dV, \quad (40)$$

the explicit nodal updates are

$$\boxed{d_I^{t+\Delta t} = d_I^t - \frac{\Delta t}{\eta_d m_I^d} F_I^d, \quad \alpha_I^{t+\Delta t} = \alpha_I^t - \frac{\Delta t}{\eta_\alpha m_I^\alpha} F_I^\alpha.} \quad (41)$$

In MPM (particle) quadrature,  $\int_{\Omega_0} (\cdot) dV \approx \sum_p (\cdot)_p V_p^0$ , so the fully discrete updates read

$$d_I^{t+\Delta t} = d_I^t - \frac{\Delta t}{\eta_d m_I^d} \sum_p \left[ \frac{g_c^d}{\ell_d} d_p^t N_I(\mathbf{x}_p) + g_c^d \ell_d \nabla N_I(\mathbf{x}_p) \cdot \nabla d^t - 2(1 - d_p^t) N_I(\mathbf{x}_p) \mathcal{H}_p \right] V_p^0, \quad (42)$$

$$\alpha_I^{t+\Delta t} = \alpha_I^t - \frac{\Delta t}{\eta_\alpha m_I^\alpha} \sum_p \left[ \frac{g_c^\alpha}{\ell_\alpha} \alpha_p^t N_I(\mathbf{x}_p) + g_c^\alpha \ell_\alpha \nabla N_I(\mathbf{x}_p) \cdot \boldsymbol{\omega}^\alpha \nabla \alpha^t - \mathcal{G}_p^\alpha N_I(\mathbf{x}_p) \right] V_p^0, \quad (43)$$

with  $d_p^t = \sum_I N_I(\mathbf{x}_p) d_I^t$ ,  $\nabla d^t = \sum_I \nabla N_I(\mathbf{x}_p) d_I^t$  (and analogously for  $\alpha$ ),  $\mathcal{H}_p$  the particle history, and  $\mathcal{G}_p^\alpha = (1 - \alpha_p^t) g(d_p^t) \boldsymbol{\varepsilon}_p : (\mathbf{T}_\sigma \mathbf{A}_2 \mathbf{T}_\sigma^T) : \boldsymbol{\varepsilon}_p$  the particle interfacial driving energy (31). After the grid updates, the particle fields are recovered by mapping back,  $d_p^{t+\Delta t} = \sum_I N_I d_I^{t+\Delta t}$ ,  $\alpha_p^{t+\Delta t} = \sum_I N_I \alpha_I^{t+\Delta t}$ , and irreversibility is enforced pointwise:  $d_p^{t+\Delta t} \leftarrow \max(d_p^t, d_p^{t+\Delta t})$ ,  $\alpha_p^{t+\Delta t} \leftarrow \max(\alpha_p^t, \alpha_p^{t+\Delta t})$ .

## 6.4 Higher-order fits of $\bar{\mathbf{C}}'(\alpha)$ enabled by the explicit scheme

In the implicit formulation of the source paper the interfacial damage problem is assembled as the linear system  $\mathbf{K}_\alpha \boldsymbol{\alpha} + \mathbf{F}_\alpha(\boldsymbol{\alpha}) = \mathbf{0}$  (its eq. 72), with  $\mathbf{K}_\alpha$  constant and

$$\mathbf{F}_\alpha(\boldsymbol{\alpha}) = \int_\Omega \frac{1}{2} g(d) \boldsymbol{\varepsilon} : \frac{\partial \bar{\mathbf{C}}^\alpha}{\partial \alpha} : \boldsymbol{\varepsilon} \mathbf{N}^T d\Omega. \quad (44)$$

This system is linear in  $\alpha$  *only if*  $\partial \bar{\mathbf{C}}^\alpha / \partial \alpha$  is affine in  $\alpha$ , which is precisely what forces the quadratic fit  $\bar{\mathbf{C}}'_{ij} = a_1^{ij} + a_2^{ij}(1 - \alpha)^2$ , so that  $\partial \bar{\mathbf{C}}'_{ij} / \partial \alpha = -2a_2^{ij}(1 - \alpha)$ . As the paper notes, the quadratic fit is accurate only for  $\chi \approx 10$ ; for other  $\chi$  a higher-order fit is needed, but that would render eq. 72 *nonlinear*, requiring Newton iteration and explicit box constraints to enforce  $0 \leq \alpha \leq 1$  — a development it defers to future work.

**The explicit scheme removes this restriction entirely.** The interfacial field is not obtained from a linear (or any) solve: it is marched by the forward–Euler update (41)/(43), in which the nodal force  $F_I^\alpha$  is *evaluated at the known state*  $\alpha^t$ . No tangent is assembled or inverted, so the dependence of  $\partial \overline{\mathbf{C}}^\alpha / \partial \alpha$  on  $\alpha$  may be of arbitrary polynomial order (indeed any smooth function): the nonlinearity is simply lagged by one step, exactly as the nonlinear terms  $g'(d)\mathcal{H}$  (and, for AT1/AT2, the crack–density derivatives) are already handled explicitly in the  $d$ –update. A general fit of order  $n_f$ ,

$$\overline{\mathbf{C}}'_{ij}(\alpha) = \sum_{k=0}^{n_f} a_k^{ij} (1 - \alpha)^k, \quad \frac{\partial \overline{\mathbf{C}}'_{ij}}{\partial \alpha} = - \sum_{k=1}^{n_f} k a_k^{ij} (1 - \alpha)^{k-1}, \quad (45)$$

may therefore be used to reproduce the homogenised curves at *any*  $\chi$  (the paper’s  $\chi = 10$ , quadratic case is  $n_f = 2$ ). The rotation is unchanged,  $\overline{\mathbf{C}}^\alpha = \mathbf{T}_\sigma \overline{\mathbf{C}}' \mathbf{T}_\sigma^T$  and  $\partial_\alpha \overline{\mathbf{C}}^\alpha = \mathbf{T}_\sigma (\partial_\alpha \overline{\mathbf{C}}') \mathbf{T}_\sigma^T$ , and the interfacial driving energy generalises to

$$\mathcal{G}^\alpha = -\frac{1}{2} g(d) \boldsymbol{\varepsilon} : \frac{\partial \overline{\mathbf{C}}^\alpha}{\partial \alpha} : \boldsymbol{\varepsilon} = \frac{1}{2} g(d) \boldsymbol{\varepsilon} : \left[ \mathbf{T}_\sigma \left( \sum_{k=1}^{n_f} k \mathbf{A}_k (1 - \alpha)^{k-1} \right) \mathbf{T}_\sigma^T \right] : \boldsymbol{\varepsilon}, \quad (46)$$

with  $\mathbf{A}_k$  the matrix of coefficients  $a_k^{ij}$ ; the quadratic case  $n_f = 2$  recovers  $\mathcal{G}^\alpha = (1 - \alpha) g(d) \boldsymbol{\varepsilon} : (\mathbf{T}_\sigma \mathbf{A}_2 \mathbf{T}_\sigma^T) : \boldsymbol{\varepsilon}$  of (31).

The box constraint  $0 \leq \alpha \leq 1$  — which in the implicit nonlinear setting would demand a constrained (bound–limited) solver — is here enforced trivially by clamping the marched value together with the irreversibility maximum,

$$\alpha_p^{t+\Delta t} \leftarrow \min \left( \max(\alpha_p^t, \alpha_{p,\text{trial}}^{t+\Delta t}), 1 \right). \quad (47)$$

The only new numerical consideration is a source–term stability limit discussed in Section 8; the diffusion and reaction operators  $\mathbf{K}_\alpha$  are unchanged by the fit order, so the anisotropic diffusion limit (51) is unaffected.

## 7 Total solution algorithm: explicit USL–MPM with two phase fields

The two damage updates are advanced inside the same P2G/G2P machinery as the momentum problem. Each material point  $p$  carries  $\mathbf{x}_p, \mathbf{v}_p, \boldsymbol{\sigma}_p, \mathbf{F}_p, V_p^0, V_p, m_p, \rho_p$ , the two damage fields  $d_p, \alpha_p$ , the history  $\mathcal{H}_p$ , and the (fixed) layer angle  $\theta$  defining  $\overline{\mathbf{C}}^\alpha(\alpha, \theta), \partial \overline{\mathbf{C}}^\alpha / \partial \alpha, \boldsymbol{\omega}^\alpha$ . In USL, stresses are updated *last*, after the grid momenta have been advanced and mapped back.

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**Algorithm 1** Explicit USL–MPM with two–field anisotropic phase–field fracture
 

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1: Initialisation: Cartesian grid;  $t \leftarrow 0$ ; precompute  $\overline{\mathbf{C}}^\alpha(\alpha, \theta)$ ,  $\partial \overline{\mathbf{C}}^\alpha / \partial \alpha$ ,  $\boldsymbol{\omega}^\alpha(\theta)$  from  $\overline{\mathbf{C}}'(\alpha)$ ,  $\mathbf{T}_\sigma(\theta)$ ,  $\mathbf{P}^\alpha$  ▷ Eqs. (14),(15),(22)
2: Particle data:  $\mathbf{x}_p^0, \mathbf{v}_p^0, \boldsymbol{\sigma}_p^0, \mathbf{F}_p^0, V_p^0, m_p, \rho_p^0, d_p^0, \alpha_p^0, \mathcal{H}_p^0$ 
3: while  $t < t_f$  do
4:   Reset grid:  $m_I, (m\mathbf{v})_I, \mathbf{f}_I^{\text{ext}}, \mathbf{f}_I^{\text{int}}, m_I^d, m_I^\alpha, F_I^d, F_I^\alpha \leftarrow 0$ 
5:   Particle  $\rightarrow$  grid (P2G)
6:      $m_I^t = \sum_p N_I m_p$ ;  $(m\mathbf{v})_I^t = \sum_p N_I (m\mathbf{v})_p^t$ 
7:      $\mathbf{f}_I^{\text{ext},t} = \sum_p N_I m_p \mathbf{b}$ ;  $\mathbf{f}_I^{\text{int},t} = -\sum_p V_p^t \boldsymbol{\sigma}_p^t \nabla N_I$  ▷  $\boldsymbol{\sigma}_p^t = g(d_p^t) \overline{\mathbf{C}}^\alpha(\alpha_p^t) : \boldsymbol{\varepsilon}_p^t$ 
8:      $\mathbf{f}_I^t = \mathbf{f}_I^{\text{ext},t} + \mathbf{f}_I^{\text{int},t}$ 
9:     Pseudo–masses:  $m_I^d = m_I^\alpha = \sum_p N_I V_p^0$ 
10:    Bulk damage force  $F_I^d = \sum_p [\frac{g_c^d}{\ell_d} d_p^t N_I + g_c^d \ell_d \nabla N_I \cdot \nabla d^t - 2(1 - d_p^t) N_I \mathcal{H}_p^t] V_p^0$ 
11:    Interf. damage force  $F_I^\alpha = \sum_p [\frac{g_c^\alpha}{\ell_\alpha} \alpha_p^t N_I + g_c^\alpha \ell_\alpha \nabla N_I \cdot \boldsymbol{\omega}^\alpha \nabla \alpha^t - \mathcal{G}_p^\alpha N_I] V_p^0$ 
12:  end
13:  Update momenta:  $(m\mathbf{v})_I^{t+\Delta t} = (m\mathbf{v})_I^t + \mathbf{f}_I^t \Delta t$ 
14:  Update bulk damage:  $d_I^{t+\Delta t} = d_I^t - \frac{\Delta t}{\eta_d m_I^d} F_I^d$ 
15:  Update interfacial damage:  $\alpha_I^{t+\Delta t} = \alpha_I^t - \frac{\Delta t}{\eta_\alpha m_I^\alpha} F_I^\alpha$ 
16:  Boundary conditions: fix  $(m\mathbf{v})_I$  on  $\partial \Omega_u$ ; prescribe  $d_I, \alpha_I$  on damage–BC nodes
17:  Grid  $\rightarrow$  particle (G2P)
18:     $\mathbf{v}_I^t = (m\mathbf{v})_I^t / m_I^t$ ,  $\mathbf{v}_I^{t+\Delta t} = (m\mathbf{v})_I^{t+\Delta t} / m_I^t$ 
19:    Particle velocity (PIC/FLIP):  $\mathbf{v}_p^{t+\Delta t} = \beta(\mathbf{v}_p^t + \sum_I N_I [\mathbf{v}_I^{t+\Delta t} - \mathbf{v}_I^t]) + (1 - \beta) \sum_I N_I \mathbf{v}_I^{t+\Delta t}$ 
20:    Particle position:  $\mathbf{x}_p^{t+\Delta t} = \mathbf{x}_p^t + \Delta t \sum_I N_I \mathbf{v}_I^{t+\Delta t}$ 
21:    Map damage:  $d_p^{t+\Delta t} = \sum_I N_I d_I^{t+\Delta t}$ ,  $\alpha_p^{t+\Delta t} = \sum_I N_I \alpha_I^{t+\Delta t}$ 
22:    Enforce irreversibility:  $d_p^{t+\Delta t} \leftarrow \max(d_p^t, d_p^{t+\Delta t})$ ,  $\alpha_p^{t+\Delta t} \leftarrow \max(\alpha_p^t, \alpha_p^{t+\Delta t})$ 
23:     $\mathbf{L}_p^{t+\Delta t} = \sum_I \nabla N_I \mathbf{v}_I^{t+\Delta t}$ ;  $\mathbf{F}_p^{t+\Delta t} = (\mathbf{I} + \mathbf{L}_p^{t+\Delta t} \Delta t) \mathbf{F}_p^t$ ;  $V_p^{t+\Delta t} = \det(\mathbf{F}_p^{t+\Delta t}) V_p^0$ 
24:     $\mathbf{D}_p^{t+\Delta t} = \frac{1}{2}(\mathbf{L}_p^{t+\Delta t} + \mathbf{L}_p^{t+\Delta t T})$ ;  $\Delta \boldsymbol{\varepsilon}_p = \Delta t \mathbf{D}_p^{t+\Delta t}$ ;  $\boldsymbol{\varepsilon}_p^{t+\Delta t} = \boldsymbol{\varepsilon}_p^t + \Delta \boldsymbol{\varepsilon}_p$ 
25:    Update stress (last):  $\boldsymbol{\sigma}_{0,p}^{t+\Delta t} = \overline{\mathbf{C}}^\alpha(\alpha_p^{t+\Delta t}, \theta) : \boldsymbol{\varepsilon}_p^{t+\Delta t}$ ;  $\boldsymbol{\sigma}_p^{t+\Delta t} = g(d_p^{t+\Delta t}) \boldsymbol{\sigma}_{0,p}^{t+\Delta t}$ 
26:    Update history:  $\psi_{0,p}^e = \frac{1}{2} \boldsymbol{\varepsilon}_p^{t+\Delta t} : \overline{\mathbf{C}}^\alpha(\alpha_p^{t+\Delta t}, \theta) : \boldsymbol{\varepsilon}_p^{t+\Delta t}$ ;  $\mathcal{H}_p^{t+\Delta t} = \max(\mathcal{H}_p^t, \psi_{0,p}^e)$ 
27:  end
28:  Advance time:  $t \leftarrow t + \Delta t$ 
29: end while

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Lines new relative to the single–field scheme are: the precomputation of the rotated tensors (line 1), the two pseudo–masses and *two* damage forces in P2G (lines 10–11), the two explicit grid updates (lines 14–15), the two particle damage / irreversibility updates in G2P (lines 21–22), and the use of the  $\alpha$ –dependent,  $\theta$ –rotated stiffness in the stress and history updates (lines 25–26).

## 8 Three explicit time–step (stability) limits

Three conditionally stable explicit updates run simultaneously; the global step must respect the smallest of the three limits,

$$\Delta t = \text{SF} \cdot \min(\Delta t_{\text{crit}}^u, \Delta t_{\text{crit}}^d, \Delta t_{\text{crit}}^\alpha), \quad 0 < \text{SF} \leq 1, \quad (48)$$

with SF a safety factor and  $h$  the background cell size.

**(1) Elastodynamic CFL — momentum update (32a).** The momentum update is limited by the Courant–Friedrichs–Lewy condition

$$\Delta t_{\text{crit}}^u = \frac{h}{c_d}, \quad c_d = \sqrt{\frac{C_{\text{max}}}{\rho}}, \quad (49)$$

where  $c_d$  is the fastest (dilatational) wave speed. For the anisotropic, damage–degraded material the conservative choice is  $C_{\text{max}} = \max_i \lambda_i(g(d)\bar{\mathbf{C}}^\alpha(\alpha, \theta))$ , the largest eigenvalue of the current stiffness matrix; for an undamaged isotropic estimate this reduces to  $c_d = \sqrt{(\lambda + 2\mu)/\rho}$ . Because  $g(d)\bar{\mathbf{C}}^\alpha \leq \bar{\mathbf{C}}^\alpha(0)$ , the intact stiffness gives a safe upper bound on  $c_d$ .

**(2) Bulk–damage diffusion limit — update (32b).** Equation (32b) is a reaction–diffusion equation with isotropic diffusivity  $D_d = g_c^d \ell_d / \eta_d$ . The forward–Euler stability of the diffusion operator on a grid of spacing  $h$  in  $n$  space dimensions requires

$$\Delta t_{\text{crit}}^d = \frac{h^2}{2nD_d} = \frac{\eta_d h^2}{2n g_c^d \ell_d}, \quad (n = 2 : \Delta t_{\text{crit}}^d = \eta_d h^2 / 4g_c^d \ell_d). \quad (50)$$

(The reaction term  $g_c^d d / \ell_d$  adds the milder restriction  $\Delta t \leq 2\eta_d \ell_d / g_c^d$ , normally non–binding since  $\ell_d \sim h$ .)

**(3) Interfacial–damage anisotropic diffusion limit — update (32c).** Equation (32c) is an *anisotropic* reaction–diffusion equation with diffusion tensor  $\mathbf{D}^\alpha = (g_c^\alpha \ell_\alpha / \eta_\alpha) \boldsymbol{\omega}^\alpha$ . Its largest eigenvalue is set by the penalised (along–layer) direction,  $\lambda_{\text{max}}(\boldsymbol{\omega}^\alpha) = 1 + \xi^\alpha$ , so the stability is governed by

$$\Delta t_{\text{crit}}^\alpha = \frac{h^2}{2nD_{\text{max}}^\alpha} = \frac{\eta_\alpha h^2}{2n g_c^\alpha \ell_\alpha (1 + \xi^\alpha)}, \quad D_{\text{max}}^\alpha = \frac{g_c^\alpha \ell_\alpha (1 + \xi^\alpha)}{\eta_\alpha}. \quad (51)$$

The factor  $(1 + \xi^\alpha)$  is the key consequence of the anisotropic gradient penalty: with  $\xi^\alpha \gg 1$  (e.g.  $\xi^\alpha = 30$ ) the interfacial update is roughly  $(1 + \xi^\alpha)$  times more restrictive than an isotropic one at equal  $\eta, g_c, \ell$ .

**Source-term (reaction) limit for higher-order fits.** When a higher-order fit (45) ( $n_f > 2$ ) is used (Section 6.4), the interfacial source  $\mathcal{G}^\alpha(\alpha)$  is nonlinear in  $\alpha$ , and the forward-Euler integration of the reaction part imposes the additional restriction

$$\Delta t \leq \frac{2\eta_\alpha}{|r_\alpha|}, \quad r_\alpha = -\frac{g_c^\alpha}{\ell_\alpha} - \frac{1}{2}g(d)\varepsilon : \frac{\partial^2 \bar{\mathbf{C}}^\alpha}{\partial \alpha^2} : \varepsilon, \quad \frac{\partial^2 \bar{\mathbf{C}}'_{ij}}{\partial \alpha^2} = \sum_{k=2}^{n_f} k(k-1)a_k^{ij}(1-\alpha)^{k-2}, \quad (52)$$

where  $r_\alpha = \partial_\alpha(\mathcal{G}^\alpha - g_c^\alpha \alpha / \ell_\alpha)$  is the local reaction rate. For the quadratic fit  $\partial^2 \bar{\mathbf{C}}^\alpha / \partial \alpha^2 = 2\mathbf{T}_\sigma \mathbf{A}_2 \mathbf{T}_\sigma^T$  is *constant*, so this limit is mild and  $\alpha$ -independent; higher orders make it  $\alpha$ -dependent but it is normally dominated by the anisotropic diffusion limit (51) (amplified by  $1 + \xi^\alpha$ ). The interfacial critical step in (48) is thus taken as  $\Delta t_{\text{crit}}^\alpha = \min(\Delta t_{\text{diff}}^\alpha, \Delta t_{\text{react}}^\alpha)$ , with the diffusion value (51) governing in practice.

**Choosing the viscosities.** As in the single-field scheme,  $\eta_d$  and  $\eta_\alpha$  are chosen large enough that the parabolic limits do not undercut the elastodynamic CFL, so the momentum condition governs and all three fields march with the same  $\Delta t$ :

$$\eta_d \gtrsim \frac{2n g_c^d \ell_d}{h^2} \Delta t_{\text{crit}}^u = \frac{2n g_c^d \ell_d}{h c_d}, \quad \eta_\alpha \gtrsim \frac{2n g_c^\alpha \ell_\alpha (1 + \xi^\alpha)}{h^2} \Delta t_{\text{crit}}^u = \frac{2n g_c^\alpha \ell_\alpha (1 + \xi^\alpha)}{h c_d}. \quad (53)$$

The viscosities must nonetheless remain small enough not to retard the physical crack speeds artificially; since  $\omega_{\text{diss}} = \eta(\dot{\cdot}) \rightarrow 0$  as the rates vanish, the converged ( $\dot{d} = \dot{\alpha} = 0$ ) crack patterns coincide with the equilibrium solutions of the original quasi-static variational model, independently of  $\eta_d, \eta_\alpha$ . Note that  $\eta_\alpha$  generally needs to be larger than  $\eta_d$  by the factor  $(1 + \xi^\alpha)$  to keep the same step.

## 9 Summary

The single-field explicit USL-MPM phase-field scheme of the companion report was extended to the anisotropic, two-field model of Li *et al.*, restricted to linear elasticity. Two damage variables are carried: an isotropic bulk layer damage  $d$  with AT2 crack density (7), and an anisotropic micro interfacial damage  $\alpha$  with the directional crack density (8) built on  $\omega^\alpha = \mathbf{I} + \xi^\alpha \mathbf{P}^\alpha$ . The homogenised interphase stiffness was taken as the fitted quadratic  $\bar{\mathbf{C}}'(\alpha)$  and rotated to global axes by the layer angle  $\theta$  through the Voigt congruence  $\bar{\mathbf{C}}^\alpha = \mathbf{T}_\sigma \bar{\mathbf{C}}' \mathbf{T}_\sigma^T$  (14), with  $\omega^\alpha(\theta)$  given in closed form by (22). Two dissipative micro-forces,  $\eta_d \dot{d}$  and  $\eta_\alpha \dot{\alpha}$ , turn the two elliptic stationarity conditions into parabolic evolution laws (32b)–(32c), each integrated by an explicit forward-Euler update on a lumped pseudo-mass (41). Bulk damage is driven by the monotone history field  $\mathcal{H} = \max \frac{1}{2} \varepsilon : \bar{\mathbf{C}}^\alpha(\alpha) : \varepsilon$ ; interfacial damage by  $\mathcal{G}^\alpha = -\frac{1}{2}g(d)\varepsilon : \partial_\alpha \bar{\mathbf{C}}^\alpha : \varepsilon$  with pointwise-max irreversibility. The whole was embedded in the explicit USL cycle (Algorithm 1) subject to the combined three-way step restriction (48), comprising the elastodynamic CFL (49), the bulk diffusion limit (50), and the anisotropy-amplified interfacial diffusion limit (51).

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